Power Plant NO_x Emissions Change Detection Using Deep Learning and TEMPO Satellite Imagery

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Abstract

NASA's Tropospheric Emissions: Monitoring of Pollution (TEMPO) instrument provides hourly daytime observations of tropospheric nitrogen dioxide (NO_2) across North America. These observations could support independent, facility-level monitoring of power plant nitrogen oxides (NO_x) emissions, filling gaps caused by continuous emissions monitoring system (CEMS) failures and facilities not subject to hourly emissions-reporting requirements. Previous studies have estimated hourly point-source emissions using empirical methods, which are highly sensitive to instrument retrieval uncertainty, changing atmospheric conditions, TEMPO's coarse spatial resolution, signals from nearby emission sources, and temporal mismatch with hourly CEMS records. We instead formulate facility monitoring as change detection over sequences of TEMPO observations. This allows the model to learn relative emissions behavior rather than estimate exact emissions from noisy data. We develop a convolutional gated recurrent unit (ConvGRU) model that processes five consecutive TEMPO observations together with meteorological, plume-transport, and tabular features encoding time of day and seasonality to classify power plant emission trajectories as increasing, steady, or decreasing. On a held-out evaluation set of more than 22,000 samples covering 116 active U.S. power plants observed between August 2023 and June 2026, the ConvGRU model achieved a macro-F1 score of 0.56, outperforming a baseline multilayer perceptron (MLP) trained only on the same time-derived features, which scored 0.45. Our findings suggest that the ConvGRU model can identify signals of power plant emissions from TEMPO observations beyond the time-derived features shared with the baseline MLP. More broadly, they indicate that change detection may offer a more tractable framework for modeling power plant activity from TEMPO observations than estimating absolute emissions.

1 Introduction

Based on U.S. Environmental Protection Agency (EPA) inventory data, power plants in the U.S. electricity-generation sector emitted approximately 0.74 million short tons of NO_x in 2023, about 9.5% of national emissions [16]. NO_x emissions contribute to ground-level ozone and secondary particulate matter, pollutants that can inflame and damage the airways, reduce lung function, and aggravate asthma [14]. EPA regulations require most large U.S. fossil-fuel-fired generating units that produce electricity for sale to submit unit-level hourly emissions data on a quarterly basis [17]. However, monitoring equipment malfunctions and maintenance can make CEMS measurements unavailable or invalid, requiring plants to report calculated substitute values instead of direct measurements [15]. Behind-the-meter generators that do not sell electricity can fall outside federal hourly reporting requirements, such as the natural gas turbines powering xAI's Colossus 2 data center [6]. Outside the United States, comparable public, hourly, unit-level emissions data are generally unavailable; global inventories typically provide only annual estimates [7]. Independent estimates could therefore fill reporting gaps and provide timely insight into facility operations ahead of the EPA's quarterly release of hourly emissions data.

TEMPO is an ultraviolet-visible spectrometer hosted aboard a geostationary satellite that observes tropospheric vertical column densities (TVCDs) for various trace gases across North America during daylight hours [19]. Earlier polar-orbiting instruments, including the Ozone Monitoring Instrument (OMI) and TROPOspheric Monitoring Instrument (TROPOMI), typically observe a given location once per day; TEMPO's geostationary view provides approximately hourly scans with finer spatial resolution [8, 18, 19]. Hourly sampling captures intraday changes in emissions plumes and atmospheric transport that daily observations can miss. However, power plant emissions signals in TEMPO NO_2 TVCDs can be weakened or obscured by factors including evolving meteorology and atmospheric chemistry, cloud cover, viewing geometry and retrieval uncertainty, dilution within TEMPO's 2.0 km × 4.75 km spatial resolution, and interference from nearby emission sources [6, 7, 13]. Cross-comparisons with CEMS data, which is the most reliable ground truth available, introduce an additional temporal mismatch because TEMPO observes a facility at a specific time during each hourly scan, whereas CEMS data aggregate emissions over an entire clock hour [13, 17].

Satellite instruments measure atmospheric column abundance rather than emissions directly. Lightweight inversion methods therefore infer source rates from the spatial structure of an observed plume and estimated winds. Cross-sectional flux (CSF) removes the background column, integrates the remaining enhancement across transects perpendicular to the plume, and multiplies the resulting line density by an effective wind speed. Under steady-state conditions, this flux approximates the source emission rate. The directional derivative approach (DDA) instead combines the wind field with spatial gradients in the observed column to estimate a local source term, which is then integrated around the facility [10, 12].

Although promising, these methods have not performed consistently across source strengths and observing conditions. In synthetic satellite scenes with realistic cloud filtering and reanalysis winds, Santaren et al. reported annual relative RMSEs of 20% for Gaussian plume inversion, 27% for CSF, and 31% for lightweight CSF [10]. Cloud filtering disproportionately reduced wintertime coverage and prevented the methods from reliably reproducing month-to-month emission variability. With TEMPO observations,

Sun et al. improved correlations with CEMS at 10 of 14 power plants by incorporating changes between consecutive NO_2 columns, although the resulting estimates remained approximately fourfold lower than CEMS measurements [12]. Beaudry et al. used a rise–decay model that separately fits near-source NO_2 buildup and downwind plume decay, estimating average emissions within 50% of CEMS values at 27 of 64 plants but exhibiting substantially greater bias below 250 kg NO_2 h^{-1} [3].

Machine learning (ML) studies have addressed several related point-source monitoring tasks. Huang and Wang trained a tabular MLP to predict whether a TROPOMI observation contained a detectable power plant NO_2 plume using 25 facility, meteorological, surface, and sensor variables sampled at the plant location [7]. Their models achieved F1 scores above 0.66, but detectability ranged from below 20% to above 60% for plants with similar emissions. Their interference filters also retained plants representing only 45.0% of U.S. and 21.1% of global NO_x emissions in their datasets. Alnaim et al. used stacked long short-term memory (LSTM) networks to predict next-day EPA-reported NO_2 emissions from seven-day sequences of previous emissions, scalar TROPOMI observations, MERRA-2 meteorology, aerosol measurements, and calendar variables [1]. Their best single-plant LSTM had an error near 40%, while multi-plant experiments exceeded 55% error and often underestimated extreme values. Dumont et al. trained CNNs to estimate power plant CO_2 emissions from simulated XCO_2 and wind fields, with optional NO_2 inputs, and obtained median relative errors of approximately 20–25% at geographically held-out plants [5]. Their evaluation used synthetic scenes that did not represent clouds or structured satellite-retrieval errors.

The point-source ML studies reviewed above do not jointly model the spatial structure and temporal evolution of consecutive trace-gas satellite rasters. We instead formulate facility monitoring as change detection over five TEMPO scans. Classifying emissions as increasing, steady, or decreasing preserves information about short-term facility behavior while reducing reliance on recovering exact emissions amounts from noisy observations.

Two relevant model families are convolutional recurrent neural networks (CRNNs) and spatiotemporal vision transformers (ViTs). At each timestep, a CRNN combines features from the current raster with a spatially structured state carried from the preceding timestep. Convolutional operations within the recurrent cell preserve the raster grid and update each location using its surrounding neighborhood. ConvLSTM controls separate cell and hidden states through input, forget, and output gates, providing explicit control over memory retention at the cost of additional parameters and computation. ConvGRU instead maintains one hidden state controlled by reset and update gates, producing a more compact recurrent model with less explicit control over long-term memory [2, 11]. Spatiotemporal ViTs divide each raster into patches and use self-attention to connect patch representations across locations and timesteps. This provides direct access to global space–time relationships without recurrent processing, but supplies a weaker locality bias and shifts more pattern learning to the training data; the computational cost also grows with the number of patches and timesteps [4].

We select ConvGRU to balance modeling capacity with parameter efficiency. Each sample spans only five scans, so ConvLSTM's additional memory state offers less apparent benefit than it would for longer sequences while increasing the number of model parameters. The input rasters are also small (24×24 pixels), and power plant plumes generally form connected clusters of pixels whose evolution can be captured through local convolutional operations rather than long-range interactions. TEMPO also provides only about three years of observations, which limits the overall dataset size. A compact ConvGRU therefore provides spatial memory without the added complexity of ConvLSTM or a spatiotemporal ViT.

2 Data Sources

2.1 TEMPO NO_2 Data

NASA distributes TEMPO NO_2 retrievals in two formats: Level 2 data on the instrument's native measurement footprints and Level 3 data regridded from each east-to-west scan [13]. We downloaded version 4 Level 2 NO_2 retrievals from August 2023 through June 2026. Along with NO_2 TVCDs, the product provides metadata including native footprint geometry, mirror-step observation times, main data quality flags, and effective cloud fractions, which we later use for regridding, quality filtering, and temporal alignment with CEMS records.

2.2 CEMS Records

We downloaded hourly emissions records for individual power plant units and annual facility and unit attributes from the U.S. Environmental Protection Agency's Clean Air Markets Program Data (CAMPD) APIs for August 2023 through June 2026. From the hourly records, we use facility and unit identifiers, reporting date and hour, NO_x mass and its measurement flag, and operating time. From the annual attributes, we use facility coordinates.

2.3 HRRR Weather Data

The High-Resolution Rapid Refresh (HRRR) is an hourly weather model with a 3 km grid covering the contiguous United States. From NOAA's public HRRR archive, we downloaded hourly 2 m air temperature and eastward and northward wind components at 80 m above ground for August 2023 through June 2026.

3 Methods

3.1 AOI Construction

For each power plant, we define a 72 km × 72 km area of interest (AOI) centered on the facility and divide it into a 24×24 grid of 3 km × 3 km cells. For each AOI-hour, we sum the unit-level NO_x mass reported by all facilities in the AOI and discard the hour if any contributing measurement is invalid or unavailable. We score each AOI using its median total NO_x mass across hours whose mean unit operating time meets or exceeds the AOI-specific median. We retain the highest-scoring 50% to focus the dataset on emissions signals more likely to be detected by TEMPO.

3.2 TEMPO Regridding and Quality Filtering

TEMPO Level 2 observations lie on irregular native footprints, whereas our model uses a fixed spatial grid. We project the footprint boundaries and AOI grid to the contiguous-U.S. Albers equal-area coordinate system. We then assign each AOI grid cell an NO_2 TVCD by averaging the TEMPO footprints that intersect it, weighted by

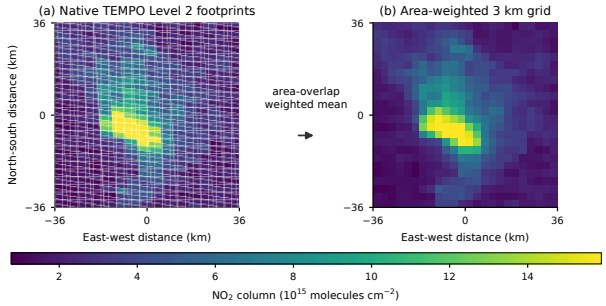


Figure 1: TEMPO Level 2 regridding to the AOI grid.

their area of overlap. Figure 1 illustrates this mapping from native footprints to the fixed AOI grid.

A footprint contributes to the NO_2 average only when a given pixel's main data quality flag equals zero (TEMPO metadata that indicates reliable retrievals) and its effective cloud fraction does not exceed 0.20. Grid cells without an accepted footprint overlap remain masked, and we retain a binary mask of the original valid cells. We use bilinear interpolation to fill masked NO_2 pixels for downstream modeling.

3.3 Sample Construction

Each sample contains five consecutive observations of one AOI, indexed from t_0 to t_4. We retain a sequence only when each pair of adjacent TEMPO scans is separated by 40–70 minutes and every NO_2 raster has valid pixel coverage of at least 85%. At each timestep, we pair the TEMPO raster with the HRRR temperature and wind rasters from the closest hour and interpolate them to the same 24×24 raster size and spatial coverage as the NO_2 rasters.

Because CEMS records correspond to clock hours rather than TEMPO scan times, we linearly interpolate the AOI-hour NO_x emissions between the surrounding hourly CEMS records at each scan time. Let E_i denote the interpolated emissions at time t_i. For scans t_a through t_b, we calculate the time-weighted mean

$$\overline{E}_{a:b} = \frac{\sum_{i=a}^{b-1}(t_{i+1} - t_i)(E_i + E_{i+1})/2}{t_b - t_a}. \tag{1}$$

We define the sequence change as $\Delta E = \overline{E}_{2:4} - \overline{E}_{0:2}$ and use the threshold $\theta = 250\ \mathrm{lb\,h^{-1}}$ to assign its category:

$$c = \begin{cases} \text{increase}, & \Delta E \geq \theta, \\ \text{steady}, & -\theta < \Delta E < \theta, \\ \text{decrease}, & \Delta E \leq -\theta. \end{cases} \tag{2}$$

Figure 2 shows one complete five-timestep model input and its derived behavior label.

3.4 Transport Model

Emissions change signals in consecutive NO_2 rasters can be degraded by movement and decay of the existing plume. For each adjacent pair, we use a simple transport model to predict the next NO_2 observation without new emissions in the AOI. We then subtract this prediction from the observed next raster to obtain a transport residual, which separates plume structure explained by modeled advection, horizontal dispersion, and chemical decay from remaining variation that may reflect emissions changes within the AOI.

For the interval $\Delta t_i = t_{i+1} - t_i$, let $N_i(\boldsymbol{x})$ and $N_{i+1}(\boldsymbol{x})$ denote the earlier and later NO_2 rasters, $\boldsymbol{u}_i(\boldsymbol{x})$ and $\boldsymbol{u}_{i+1}(\boldsymbol{x})$ their wind fields, and B_i the median NO_2 value along the boundary of the earlier raster. Using these quantities, we construct the transported prediction and residual as follows:

$$\text{(i)} \quad \overline{\boldsymbol{u}}_i = \gamma[(1-\lambda)\boldsymbol{u}_i + \lambda \boldsymbol{u}_{i+1}]$$

$$\text{(ii)} \quad A_i(\boldsymbol{x}) = N_i(\boldsymbol{x} - \overline{\boldsymbol{u}}_i(\boldsymbol{x})\Delta t_i)$$

$$\text{(iii)} \quad \sigma_i = \sigma_0\sqrt{\frac{\Delta t_i}{1\,\mathrm{h}}}$$

$$\text{(iv)} \quad D_i = B_i + G_{\sigma_i} * (A_i - B_i)$$

$$\text{(v)} \quad \widehat{N}_{i+1} = B_i + \exp\left(-\frac{\Delta t_i}{\tau}\right)(D_i - B_i)$$

$$\text{(vi)} \quad R_i = N_{i+1} - \widehat{N}_{i+1}.$$

Equations (i) and (ii) represent wind-driven advection. Satellite columns do not resolve the vertical distribution of plume mass, so point-source methods commonly use an effective wind to approximate its mean transport speed [9, 10]. We approximate this wind by averaging the fields at the two endpoints ($\lambda = 0.5$) and applying a modest speed correction ($\gamma = 0.8$). Equation (ii) then applies semi-Lagrangian advection by tracing each destination cell backward along the effective wind over Δt_i and bilinearly sampling the earlier raster at the resulting departure location.

Equations (iii) and (iv) approximate unresolved turbulent horizontal dispersion. Gaussian plume representations are widely used in satellite point-source studies [9, 10]. We set σ_0 to one 3 km grid cell after one hour and increase the width with $\sqrt{\Delta t_i}$. Applying the Gaussian kernel to $A_i - B_i$ spreads the plume enhancement while preserving the estimated background.

Equation (v) applies first-order loss to the above-background enhancement. The effective timescale $\tau = 8$ h represents the combined persistence of the observed NO_2 plume under chemical removal and other unresolved losses. Exponential loss is commonly used to describe the downwind decay of satellite-observed NO_x plumes, whose effective lifetimes vary with meteorology and plume chemistry [9]. Equation (vi) subtracts the resulting transport-only prediction from the next observation.

Each five-observation sample yields four residual rasters. We apply the same advection and diffusion operations to the earlier validity mask and combine it with the later mask to produce quality masks for the transport residuals. The transport-residual panel in Figure 2 shows the four adjacent-scan residuals for the same example sequence.

3.5 Dataset Stratification

Our split design has three goals: prevent leakage from overlapping AOIs, produce an approximately 70%/15%/15% stratification of samples, and construct class distributions that support balanced training and comparable evaluation across the held-out splits. We group overlapping AOIs into connected clusters and treat each

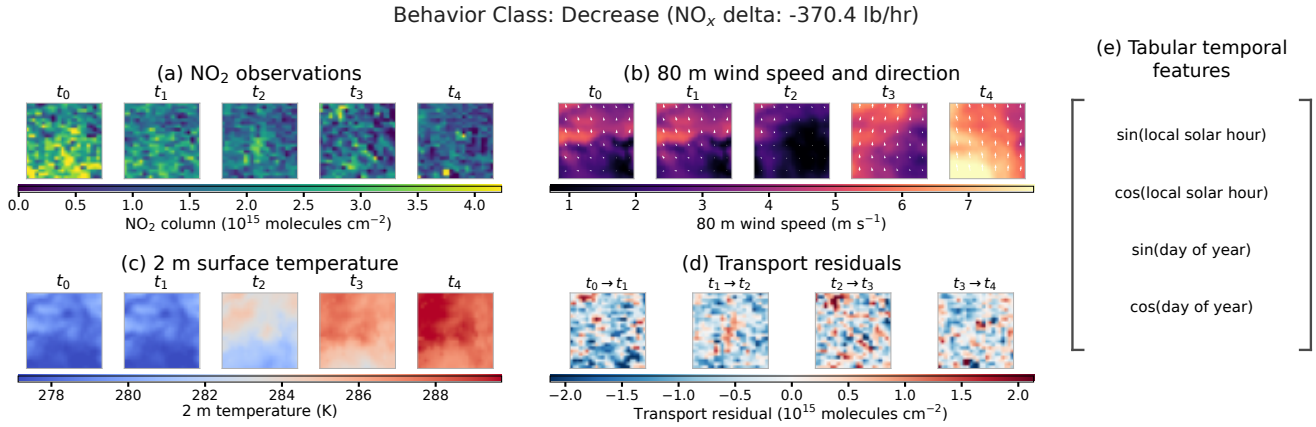


Figure 2: Complete model inputs and behavior label for one five-timestep training sample.

cluster as an indivisible assignment unit. Because each cluster contains an arbitrary number of AOIs, samples, and class instances, cluster-level assignment cannot generally meet every target exactly.

We use mixed-integer linear programming (MILP) to assign each cluster to exactly one split while minimizing normalized deviations from target AOI and eligible-sample counts. The first optimization targets a 70%/15%/15% sample stratification using AOI and sample counts, after which we fix the training clusters. A second optimization assigns the remaining clusters equally between validation and test while balancing AOI, sample, and per-class counts. It also constrains both held-out splits to have the same median number of plant units per AOI as the combined held-out pool.

We then uniformly downsample each training class to the size of the smallest class. In validation and test, we retain every increase and decrease sample and uniformly subsample steady samples to at most the size of the larger non-steady class. Subsequent raster-quality filtering can alter the final class counts, and we do not rebalance the surviving samples.

3.6 ConvGRU Model

The ConvGRU model encodes five observation frames and four interval-aligned transport residuals using branch-specific convolutional encoders whose weights are shared across time. It updates a spatial hidden state over four recurrent transitions, then combines the final visual representation with time-derived tabular features for classification. Figure 3 summarizes the complete architecture and one ConvGRU cell.

Each observation frame contains five channels: NO_2, its quality mask, the eastward and northward 80 m wind components, and 2 m temperature. Each transport-residual frame contains the residual and its quality mask. Separate observation and residual encoders account for the different meanings of these inputs. We denote them E_x and E_r, respectively, in Figure 3(a). The same encoder weights process every frame within a branch. Both encoders begin with a 5×5 convolution that produces 32 channels, followed by five residual blocks. Strided blocks increase the channel width from 32 to 48 and then 64 while reducing each 24×24 raster to a $64 \times 6 \times 6$ feature map. Each residual block applies two 3×3 convolutions with group normalization, a SiLU activation after the first convolution, and another SiLU activation after adding the skip connection. Blocks that change channel width or spatial resolution use a 1×1 convolution on the skip path.

As shown in Figure 3(a), the encoded t_0 observation initializes the hidden state, so the model does not synthesize a residual before the first observation. At each subsequent timestep t_i, the model concatenates the encoded observation with the encoded residual for $t_{i-1} \to t_i$. A 1×1 projection fuses the two encoded feature maps before the recurrent update.

Figure 3(b) expands one recurrent update. Let $\boldsymbol{f}_i$ denote the fused input and $\boldsymbol{h}_{i-1}$ the previous hidden state, a $64 \times 6 \times 6$ feature map that carries spatial information from earlier observations. The ConvGRU computes

$$[\boldsymbol{r}_i, \boldsymbol{z}_i] = \sigma\big(W_g * [\boldsymbol{f}_i, \boldsymbol{h}_{i-1}] + \boldsymbol{b}_g\big), \tag{3}$$

$$\widetilde{\boldsymbol{h}}_i = \tanh\big(W_h * [\boldsymbol{f}_i, \boldsymbol{r}_i \odot \boldsymbol{h}_{i-1}] + \boldsymbol{b}_h\big), \tag{4}$$

$$\boldsymbol{h}_i = (1 - \boldsymbol{z}_i) \odot \boldsymbol{h}_{i-1} + \boldsymbol{z}_i \odot \widetilde{\boldsymbol{h}}_i, \tag{5}$$

where the subscripts g and h denote the gate and candidate-state convolutions, respectively. W_g and W_h are their learned 3×3 kernels, while $\boldsymbol{b}_g$ and $\boldsymbol{b}_h$ are the corresponding biases. The symbol $*$ denotes convolution, and $[\cdot, \cdot]$ denotes channel-wise concatenation. The reset gate $\boldsymbol{r}_i$ controls how much previous spatial information contributes to the candidate state, while the update gate $\boldsymbol{z}_i$ determines how much of that candidate replaces the previous state.

After the fourth recurrent update, global average and maximum pooling summarize distributed and localized evidence, respectively, in the final hidden state. The model concatenates their outputs and projects them into a visual embedding. The tabular branch receives four cyclic features computed at the sequence midpoint: the sine and cosine of local solar hour and the sine and cosine of day of year. A feedforward encoder maps them into a tabular embedding. The model introduces this embedding only after recurrent processing, a late-fusion design that keeps the tabular features out of the spatial state updates. The fused representation passes through the classification head to produce logits for the three behavior classes.

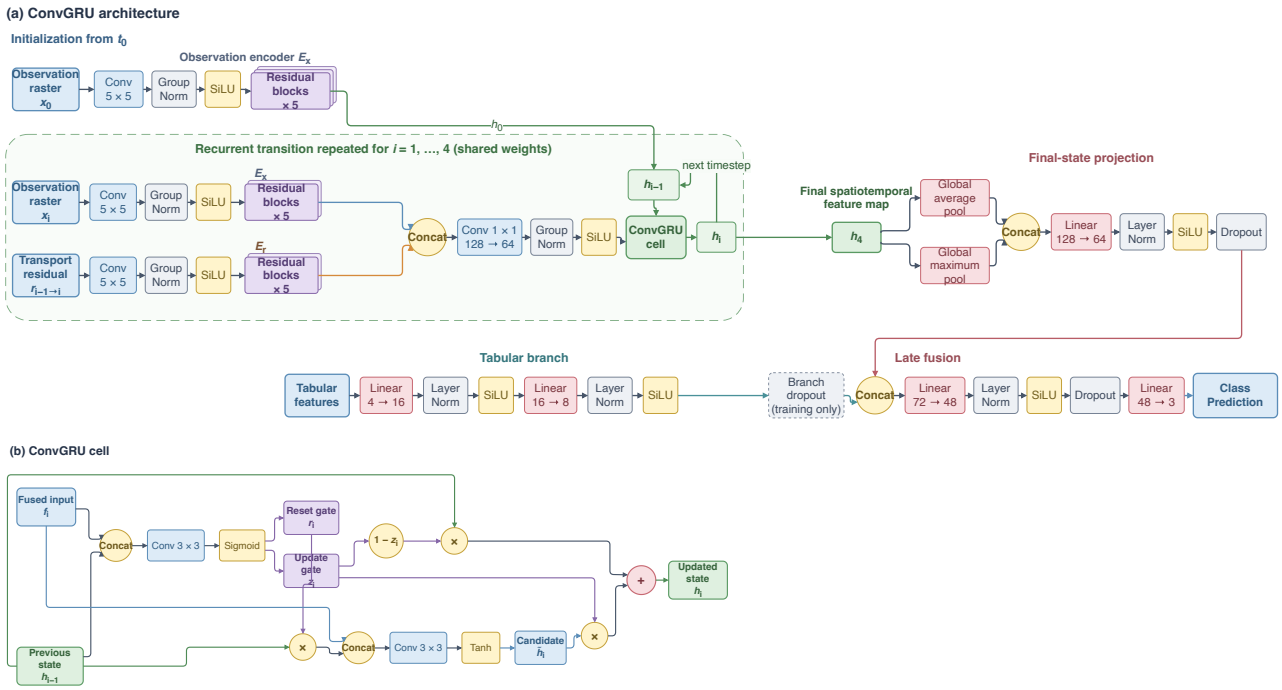


Figure 3: (a) ConvGRU high-level architecture. (b) ConvGRU cell design.

Table 1: Dataset composition by split and class.

Split	AOIs	Active plants	Decrease	Steady	Increase	Total
Training	469	530	17,130	23,289	29,888	70,307
Validation	100	109	3,240	7,648	9,902	20,790
Test	98	116	3,142	8,722	10,602	22,466
Total	667	755	23,512	39,659	50,392	113,563

Table 2: Training configuration.

Parameter	Baseline MLP	ConvGRU
Trainable parameters	291	724,939
Optimizer	AdamW	
Batch size	128	128
Initial learning rate	10^{-3}	6.04×10^{-5}
Weight decay	10^{-4}	10^{-4}
Activation dropout	–	0.20
Tabular branch dropout	–	0.25
Gradient-norm limit	5.0	
LR scheduler	Reduce on plateau	
LR reduction factor	0.5	
Scheduler patience	5 epochs	
Selection metric	Validation macro-F1	
Early-stopping patience	10 epochs	
Hardware	1 NVIDIA RTX A5000 GPU	

3.7 Baseline MLP Model

We use a lightweight baseline MLP with time-derived features to gauge how much emissions change behavior can be predicted from recurring daily and annual patterns without satellite or weather inputs. We represent each sample using four cyclic features computed at the sequence midpoint: the sine and cosine of local solar hour and the sine and cosine of day of year. The baseline MLP maps these features through 16- and 8-unit hidden layers, each with layer normalization and a SiLU activation, followed by a linear layer that produces logits for the three behavior classes.

3.8 Training Procedure

Table 1 summarizes the final dataset used to train and evaluate both models. We standardize the NO_2, wind, temperature, and transport residual rasters and the tabular inputs in every split using means and standard deviations computed from the training split.

For the ConvGRU model, we also apply tabular branch dropout during training: for each sample, the tabular embedding is fully zeroed with probability 0.25 before late fusion. This discourages the classifier from relying on the time-derived tabular features alone.

Both classifiers use inverse-frequency class weights computed from the final training split and minimize weighted cross-entropy.

4 Results

4.1 Classification Performance

We evaluate classification using macro-F1 and balanced accuracy to account for class imbalance. On the test set, the ConvGRU model achieves a macro-F1 of 0.559 and balanced accuracy of 0.563, compared with 0.454 and 0.481 for the baseline MLP. As shown in Figure 4, the ConvGRU model demonstrates improved per-class F1

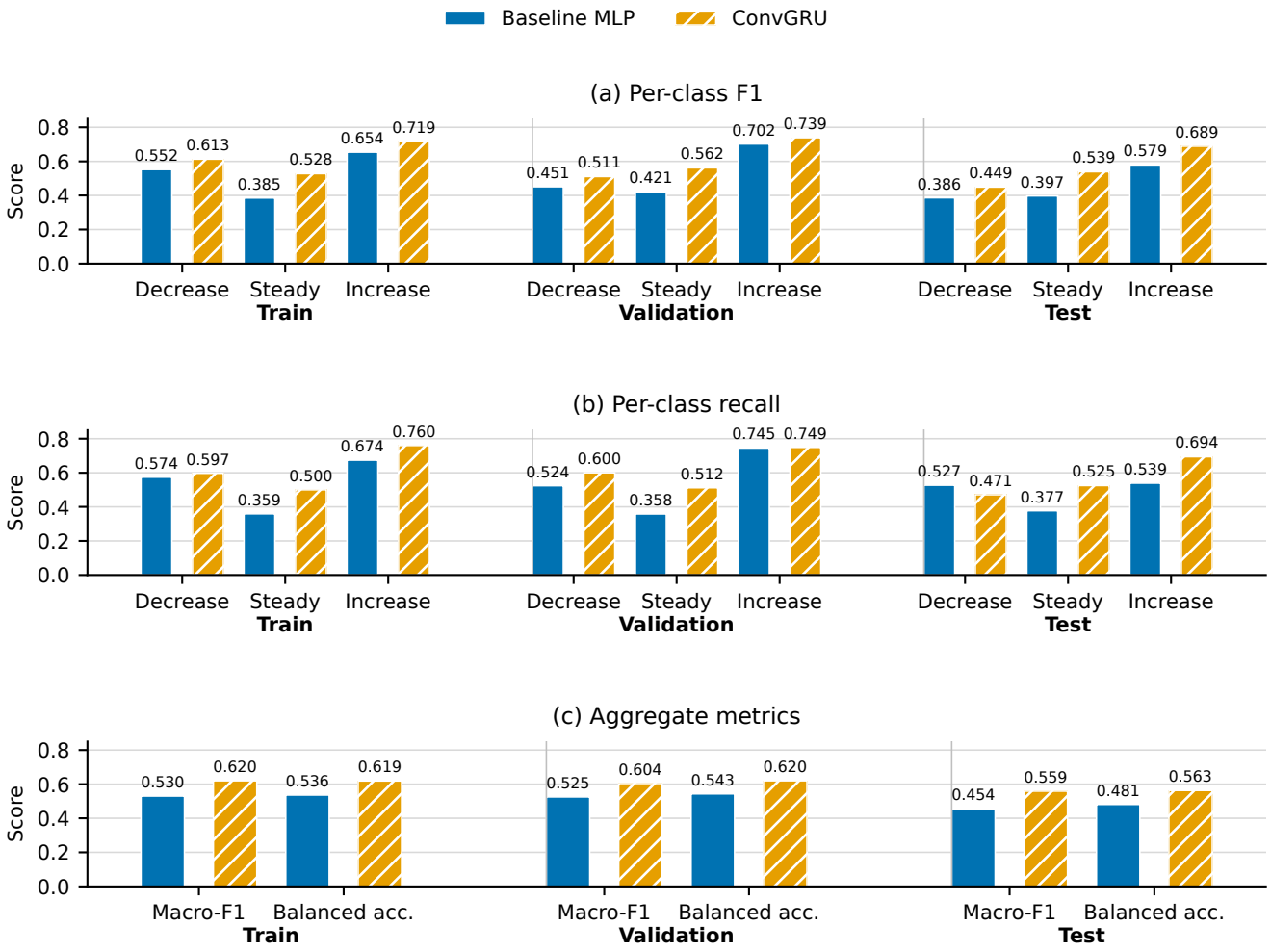


Figure 4: Classification performance by split.

and recall scores over the baseline MLP across nearly all per-split evaluations.

The largest performance gap between the models consistently occurs for steady samples, ranging from 0.141 to 0.143 F1 points across splits. This makes intuitive sense because steady samples are less likely to belong to a distinct season (e.g., increases occur more often during summer and decreases occur more often during winter), which means time-only features explain less variation in plant emissions activity for those samples. The ConvGRU model shows its weakest performance on decrease samples, with a test recall of 0.471, below the baseline MLP's 0.527. The weaker performance on decrease samples may have two causes. First, decrease events are less prevalent than increase events in the dataset, giving the model fewer examples from which to learn. Second, after facility emissions decline, previously emitted NO_2 can remain visible as the plume advects, disperses, and decays, delaying the raster response to the underlying activity change.

4.2 Performance Across Evaluation Strata

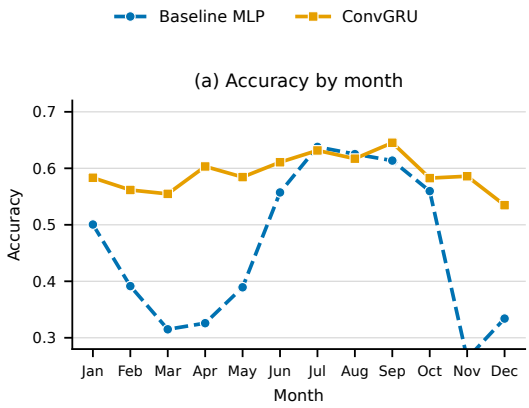


Figure 5: Classification accuracy by month.

4.2.1 Performance Across Months. As shown in Figure 5, the ConvGRU model shows more consistent monthly accuracy than the baseline MLP. The models perform most similarly during summer, while their accuracy gap reaches 0.277 in April and 0.323 in November. These results are consistent with our observation that increase samples occur more often during summer and decrease samples occur more often during winter. While the baseline MLP is restricted to learning seasonal patterns for power plant activity only, the ConvGRU model can use visual signal to predict emissions behavior when seasonality alone is generally a poor indicator of activity, such as during spring and fall.

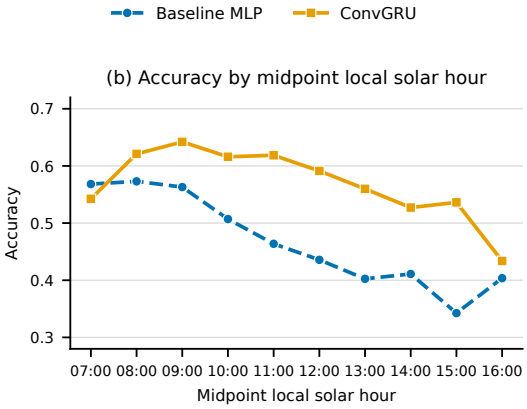


Figure 6: Classification accuracy by midpoint local solar hour.

4.2.2 Performance by Local Solar Hour. As shown in Figure 6, accuracy generally declines from morning to afternoon for both models, although the ConvGRU model outperforms the baseline MLP over nearly all hours. This pattern is consistent with changes in class composition across the day: increase samples are more prevalent earlier, whereas decrease samples, on which both models perform less accurately, become more prevalent later. Because accuracy depends on class prevalence, these results do not isolate an independent effect of solar hour.

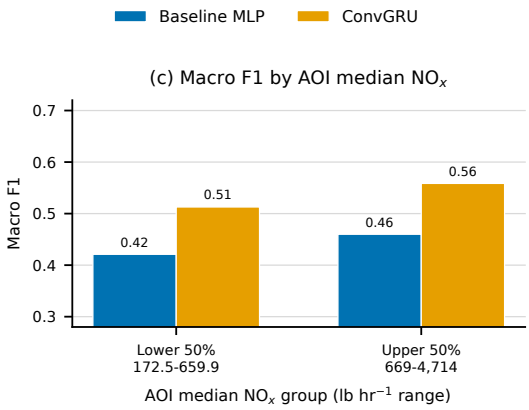


Figure 7: Macro-F1 by AOI median NO_x emissions group.

4.2.3 Performance by AOI Median NO_x Emissions. As shown in Figure 7, macro-F1 increases from 0.42 to 0.46 for the baseline MLP and from 0.51 to 0.56 for the ConvGRU model between the lower and upper halves of test AOIs by median NO_x emissions. Because both models improve similarly, higher-emission AOIs appear to constitute an easier prediction subset rather than providing an advantage attributable only to stronger raster signals. The ConvGRU model retains a nearly constant advantage over the baseline in both groups, indicating that the raster inputs provide additional information across the observed emissions range. Class composition, operating regularity, and source density may also contribute to the difference between groups.

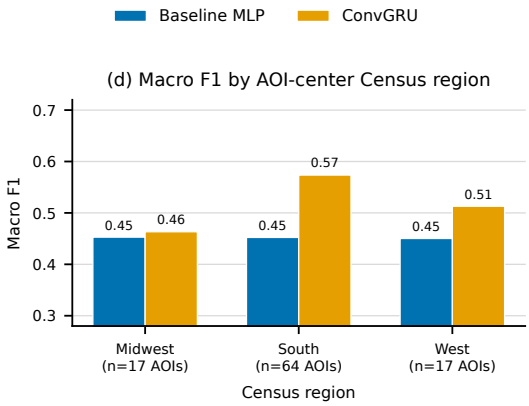


Figure 8: Macro-F1 by AOI-center Census region.

4.2.4 Performance by Census Region. As shown in Figure 8, the ConvGRU model reaches macro-F1 scores of 0.57 in the South, 0.51 in the West, and 0.46 in the Midwest, while the baseline MLP's performance remains near 0.45 across all three regions. Regional class composition may contribute to these differences. Increase samples comprise 50.5% of Southern test records, the Midwest has the largest decrease share at 26.1%, and steady samples comprise 58.9% of Western records.

AOI source density follows the same ordering as the ConvGRU model's performance. Southern test AOIs average 5.05 active plants and 17.92 units, compared with 4.00 plants and 13.59 units in the West and 2.71 plants and 6.71 units in the Midwest. The Midwest also shows the largest shift from its training distribution, where AOIs average 4.29 active plants and 13.10 units. Source density and similarity to the training distribution may therefore affect the strength and generalizability of the visual emissions patterns available to the ConvGRU model.

Training representation may also matter. Southern records comprise 62.8% of the training set, compared with 21.3% from the Midwest and 5.1% from the West. Class composition, source density, fuel mix, facility design, and operating practices vary together, so this aggregate comparison cannot isolate their individual effects. We exclude the Northeast because the test split contains no Northeastern AOIs.

4.3 Ablation Study

We retrain the ConvGRU model for each ablation in Table 3, where feature removal refers to fully zeroing out a given feature for all dataset samples. For a timestep ablation, we remove the NO_2 and weather rasters at that timestep but retain the transport residual input that overlaps it. We exclude a t_0 removal ablation because it initializes the hidden state for the ConvGRU model and therefore cannot be compared in the same manner as the other timesteps.

Ablation results show that removing the time-derived tabular features or wind rasters reduced test macro-F1 by only 0.74 and 0.27 percentage points, respectively, indicating limited marginal value when the other inputs were available. Removing the transport residuals produced the largest decline of 12.58 percentage points, suggesting that the model relies primarily on spatiotemporal raster information and benefits from a physics-guided representation that distinguishes observed changes from predicted plume transport, dispersion, and decay. Removing the temperature rasters reduced macro-F1 by 4.87 percentage points, substantially more than removing wind. Temperature may indirectly capture variation in electricity demand and plant activity, although it may also encode meteorological conditions affecting plume evolution and NO_2 retrieval.

Table 3: Ablation study on the test split.

Input feature removed	Test macro-F1	Test balanced accuracy	Test decrease F1	Test steady F1	Test increase F1
None (baseline)	0.5582	0.5667	0.4614	0.5233	0.6901
Wind rasters	0.5555	0.5661	0.4577	0.5407	0.6683
Temperature rasters	0.5095	0.5265	0.4217	0.5349	0.5719
Transport residuals	0.4324	0.4859	0.3640	0.3088	0.6245
Time-derived tabular features	0.5508	0.5673	0.4634	0.5080	0.6808
t_1 observation	0.5423	0.5379	0.4421	0.5616	0.6232
t_2 observation	0.5439	0.5562	0.4666	0.5404	0.6248
t_3 observation	0.5253	0.5274	0.4487	0.4965	0.6309
t_4 observation	0.5374	0.5406	0.4657	0.5108	0.6359

5 Discussion

The ConvGRU model's improvement over the baseline MLP indicates that consecutive TEMPO rasters contain information associated with facility activity beyond recurring daily and seasonal patterns.

Performance differences across classes provide insight into the task. The largest gain over the baseline occurred for steady classifications, for which time-derived patterns offered limited discrimination. Decreases remained more difficult to identify than increases. Residual plume mass can persist after emissions decline, creating a lag between the CEMS-derived label and the observed NO_2 field. Differences across months and solar hours may also reflect changes in class prevalence, cloud coverage, retrieval quality, and atmospheric lifetime. Regional variation could arise from differences in emissions magnitude, fuel mix, operating practices, or sample composition. These factors remain confounded in the current analysis and require controlled comparisons before assigning a causal explanation.

The ablations support the value of physics-guided preprocessing. Transport residuals may help expose changes that raw raster sequences obscure through plume motion and decay. The small effect of removing wind rasters does not imply that wind is uninformative because the transport residuals already encode wind-derived information. Temperature may capture both plume behavior and demand-related operating patterns, so its contribution cannot yet be attributed to one mechanism.

Several limitations constrain interpretation. TEMPO measures NO_2 columns, whereas CEMS reports stack-level NO_x emissions. Retrieval uncertainty, masked pixel interpolation, simplified transport assumptions, and temporal mismatch can weaken their correspondence. Because an AOI may contain multiple power plants, its label represents their aggregate emissions change and cannot identify which facility produced the observed signal. The short TEMPO record and restriction to U.S. power plants limit evidence for transfer across years, regions, and source types.

Future studies should reevaluate the method as more TEMPO observations become available and as retrieval algorithms improve. Future instruments with finer spatial resolution may provide another opportunity for evaluation. More granular ablations and controlled analyses could separate the effects of retrieval quality, meteorology, plant characteristics, and class prevalence. Sensitivity analyses should also vary the NO_x change thresholds used to construct the behavior labels. Model comparisons could include ConvLSTM and other spatiotemporal architectures, while sequence-length experiments could test whether fewer scans suffice or longer histories better capture plume evolution. Finally, future versions could replace the current transport calculation with a more complete treatment of advection, dispersion, chemistry, and plume injection height. Researchers could tune and validate its parameters using observed sequences in which CEMS reports near-zero emissions, allowing them to evaluate the transported prediction with little contribution from newly emitted NO_x.

6 Conclusion

Our results show that sequences of TEMPO observations contain information about short-term power plant activity beyond recurring daily and seasonal operating patterns. The ConvGRU model improved upon the baseline MLP across most evaluations, while the ablation study identified physics-guided transport residuals as an important source of predictive information. Together, these findings support change detection as a promising way to extract AOI-level emissions change signals from noisy satellite observations without requiring exact hourly emissions estimates.

This approach does not replace direct emissions measurements or resolve the uncertainties involved in attributing observed NO_2 columns to individual facilities. It could, however, complement existing monitoring systems by screening for short-term emissions changes in power plant regions where timely facility-level data are unavailable. Additional years of TEMPO observations, improved retrievals, and validation across regions and source types will determine how broadly the method can support independent emissions monitoring.

References

[1] Ahmed Alnaim, Ziheng Sun, and Daniel Tong. 2022. Evaluating Machine Learning and Remote Sensing in Monitoring NO_2 Emission of Power Plants. *Remote Sensing* 14, 3 (2022), 729. doi:10.3390/rs14030729

[2] Nicolas Ballas, Li Yao, Chris Pal, and Aaron Courville. 2015. Delving Deeper into Convolutional Networks for Learning Video Representations. arXiv:1511.06432 [cs.LG] doi:10.48550/arXiv.1511.06432

[3] Samuel James Beaudry, Qindan Zhu, and Ronald C. Cohen. 2026. Quantification of Power Plant NO_x Emissions: A Reference for Diurnal and Seasonal Accuracy of TEMPO. doi:10.31223/X5B808 EarthArXiv preprint.

[4] Gedas Bertasius, Heng Wang, and Lorenzo Torresani. 2021. Is Space-Time Attention All You Need for Video Understanding?. In *Proceedings of the 38th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 139)*. PMLR, Virtual Event, 813–824.

[5] J. Dumont Le Brazidec, P. Vanderbecken, A. Farchi, G. Broquet, G. Kuhlmann, and M. Bocquet. 2024. Deep Learning Applied to CO_2 Power Plant Emissions Quantification Using Simulated Satellite Images. *Geoscientific Model Development* 17 (2024), 1995–2014. doi:10.5194/gmd-17-1995-2024

[6] Kevin D. Gauld, Daniel J. Varon, Nicholas Balasus, and Daniel H. Cusworth. 2026. Quantifying AI Data Center Nitrogen Oxide (NO_x) Emissions from Space. arXiv:2608.22153 [physics.ao-ph] doi:10.48550/arXiv.2608.22153

[7] Ruizhe Huang and Sherrie Wang. 2026. Global Variability in the Detectability of Power Plant NO_2 Plumes from Space. *Atmospheric Measurement Techniques* 19 (2026), 6099–6124. doi:10.5194/amt-19-6099-2026

[8] Pieternel F. Levelt, Gijsbertus H. J. van den Oord, Marcel R. Dobber, Anssi Mälkki, Huib Visser, Johan de Vries, Piet Stammes, Jens O. V. Lundell, and Heikki Saari. 2006. The Ozone Monitoring Instrument. *IEEE Transactions on Geoscience and Remote Sensing* 44, 5 (2006), 1093–1101. doi:10.1109/TGRS.2006.872333

[9] Samuel Meier, E. F. M. Koene, Maarten Krol, Dominik Brunner, Alexander Damm, and Gerrit Kuhlmann. 2024. A Lightweight NO_2-to-NO_x Conversion Model for Quantifying NO_x Emissions of Point Sources from NO_2 Satellite Observations. *Atmospheric Chemistry and Physics* 24 (2024), 7667–7686. doi:10.5194/acp-24-7667-2024

[10] D. Santaren, J. Hakkarainen, G. Kuhlmann, E. Koene, F. Chevallier, I. Ialongo, H. Lindqvist, J. Nurmela, J. Tamminen, L. Amorós, D. Brunner, and G. Broquet. 2025. Benchmarking Data-Driven Inversion Methods for the Estimation of Local CO_2 Emissions from Synthetic Satellite Images of XCO_2 and NO_2. *Atmospheric Measurement Techniques* 18 (2025), 211–239. doi:10.5194/amt-18-211-2025

[11] Xingjian Shi, Zhourong Chen, Hao Wang, Dit-Yan Yeung, Wai-Kin Wong, and Wang-Chun Woo. 2015. Convolutional LSTM Network: A Machine Learning Approach for Precipitation Nowcasting. In *Advances in Neural Information Processing Systems*, Vol. 28. Curran Associates, Inc., Montreal, Canada, 802–810.

[12] Kang Sun, Jobaer Ahmed Saju, Caroline R. Nowlan, Gonzalo González Abad, and Xiong Liu. 2025. Hourly Nitrogen Oxides Emissions Estimated From TEMPO and Comparison With Facility-Level Monitoring Data. *Journal of Geophysical Research: Atmospheres* 130, 23 (2025), e2025JD044565. doi:10.1029/2025JD044565

[13] TEMPO Science Team. 2026. *TEMPO Trace Gas and Cloud Level 2 and 3 Data Products: User Guide.* NASA Langley Atmospheric Science Data Center. doi:10.5067/doc/tempo/L2andL3UserGuide/V2.1 Version 2.1.

[14] U.S. Environmental Protection Agency. 2016. *Integrated Science Assessment for Oxides of Nitrogen—Health Criteria.* Technical Report EPA/600/R-15/068. U.S. Environmental Protection Agency. https://www.epa.gov/isa/integrated-science-assessment-isa-oxides-nitrogen

[15] U.S. Environmental Protection Agency. 2021. *Substitute Data in EPA CAMD's Power Sector Emissions Data.* Technical Report. U.S. Environmental Protection Agency. https://www.epa.gov/system/files/documents/2021-06/monitoring-insights-substitute-data.pdf

[16] U.S. Environmental Protection Agency. 2026. Air Pollutant Emissions Trends Data. https://www.epa.gov/air-emissions-inventories/air-pollutant-emissions-trends-data Accessed October 4, 2026.

[17] U.S. Environmental Protection Agency. 2026. Emissions Monitoring and Reporting. https://www.epa.gov/power-sector/emissions-monitoring-and-reporting Accessed October 4, 2026.

[18] J. P. Veefkind, I. Aben, K. McMullan, H. Förster, J. de Vries, G. Otter, J. Claas, H. J. Eskes, J. F. de Haan, Q. Kleipool, M. van Weele, O. Hasekamp, R. Hoogeveen, J. Landgraf, R. Snel, P. Tol, P. Ingmann, R. Voors, B. Kruizinga, R. Vink, H. Visser, and P. F. Levelt. 2012. TROPOMI on the ESA Sentinel-5 Precursor: A GMES Mission for Global Observations of the Atmospheric Composition for Climate, Air Quality and Ozone Layer Applications. *Remote Sensing of Environment* 120 (2012), 70–83. doi:10.1016/j.rse.2011.09.027

[19] Peter Zoogman, Xiong Liu, R. M. Suleiman, W. F. Pennington, David E. Flittner, Jassim A. Al-Saadi, Brian B. Hilton, Dennis K. Nicks, Michael J. Newchurch, J. L. Carr, Scott J. Janz, M. R. Andraschko, Antti Arola, Barry D. Baker, Brian P. Canova, Christopher Chan Miller, Ronald C. Cohen, J. E. Davis, M. E. Dussault, David P. Edwards, Jack Fishman, A. Ghulam, Gonzalo González Abad, Michel Grutter, Jay R. Herman, J. Houck, Daniel J. Jacob, Joanna Joiner, Brian J. Kerridge, Jhoon Kim, Nickolay A. Krotkov, Lok N. Lamsal, Can Li, Anders Lindfors, Randall V. Martin, C. Thomas McElroy, Chris McLinden, Vijay Natraj, Daniel O. Neil, Caroline R. Nowlan, E. J. O'Sullivan, Paul I. Palmer, R. Bradley Pierce, Mark R. Pippin, Alfonso Saiz-Lopez, Robert J. D. Spurr, James J. Szykman, Omar Torres, J. Pepijn Veefkind, Ben Veihelmann, Hui Wang, Jun Wang, and Kelly Chance. 2017. Tropospheric Emissions: Monitoring of Pollution (TEMPO). *Journal of Quantitative Spectroscopy and Radiative Transfer* 186 (2017), 17–39. doi:10.1016/j.jqsrt.2016.05.008